

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	4	2	3	2	2	1	3	2	4	3	1	4	4	3	4	4	1	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	2	4	2	3	1	3	1	3	2	1	2	3	1	3	3	3	3	1
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	3	1	2	2	2	3	1	2	3	2	1	4	2	4	3	2	2	1	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	4	4	1	2	4	4	2	3	1	3	3	4	4	1	4	2	3	3	4	2
Q.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	2	2	2	3	4	2	2	2	4	2	1	1	1	2	1	1	2	3	3	2
Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	3	3	2	3	4	4	4	3	2	3	1	1	2	1	2	4	1	3	2
Q.	121	122	123	124	125	126	127	128	129	130										
A.	4	2	2	1	4	3	3	1	2	1										

9610ZBS107116230010

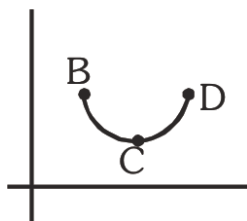
Your Hard Work Leads To Strong Foundation

Open

HINT – SHEET

1. Ans (3)

Continuously increasing slope.



2. Ans (4)

$$V_{20\Omega} = V_{\text{Total}}$$

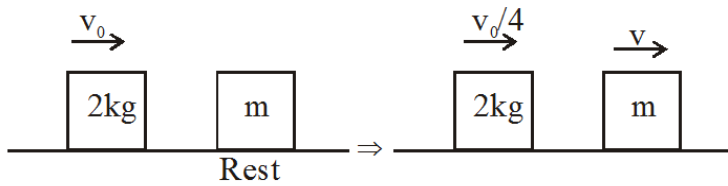
$$\therefore (20)(0.3) = (R_{\text{Total}})(0.8)$$

$$\therefore R_{\text{Total}} = \frac{30}{4}\Omega$$

$$\therefore \frac{4}{30} = \frac{1}{R_1} + \frac{1}{20} + \frac{1}{15}$$

Solving we get $R_1 = 60\Omega$

3. **Ans (2)**



By conservation of linear momentum :-

$$2v_0 = 2\left(\frac{v_0}{4}\right) + mv \Rightarrow 2v_0 = \frac{v_0}{2} + mv$$

$$\Rightarrow \frac{3v_0}{2} = mv \quad \dots(1)$$

Since collision is elastic $\rightarrow$

$$V_{\text{separation}} = V_{\text{approach}}$$

$$\Rightarrow v - \frac{v_0}{4} = v_0$$

$$\Rightarrow \frac{5v_0}{4} = v \quad \dots(2)$$

equating (2) and (1)

$$\frac{3v_0}{2} = m\left(\frac{5v_0}{4}\right)$$

4. **Ans (3)**

When key K is kept closed, condenser C is charge to potential V. When plates of capacitor are moved apart, its capacitance, $C = \frac{\epsilon_0 A}{d}$ decreases.

As potential of condenser remains same, charge $Q = CV$ decreases. So option (3) is correct.

Once key K is closed, condenser gets charged, $Q = CV$

Now, if key K is opened, battery is disconnected, no more charging can occur i.e. Q remains same.

As plates of capacitor are moved apart, its capacity $C = \frac{\epsilon_0 A}{d}$ decreases.

Therefore, its potential, $V = \frac{q}{C}$ increases.

5. **Ans (2)**

$$T = m(\ell + x)\omega^2$$

$$Kx = m(\ell + x)\omega^2$$

6. **Ans (2)**

$$\Delta L = \alpha L \Delta T = \frac{FL}{AY}$$

$$\Rightarrow \text{Stress} = \frac{F}{A} = Y \alpha \Delta T$$

7. **Ans (1)**

$t = 0$, $L \rightarrow$ open CKT

$t = \infty$, $L \rightarrow$ conductor

8. **Ans (3)**

$$P_{av} = V_{rms} i_{rms} \cos \phi$$

9. **Ans (2)**

$$H = I^2 RT$$

$$\frac{\Delta H}{H} \times 100 = \pm 2 \frac{\Delta I}{I} \times 100 \pm \frac{\Delta R}{R} \times 100 \pm \frac{\Delta t}{t} \times 100$$

$$= \pm 2 \times 3\% \pm 4\% \pm 6\% = \pm 16\%$$

10. **Ans (4)**

$$|Q_1| > |Q_2|$$

$$Q_1 \rightarrow -ve$$

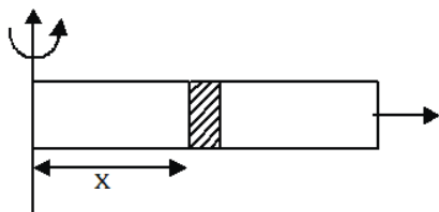
$$Q_2 \rightarrow +ve$$

Electric field will be zero, right side of Q_2

11. **Ans (3)**

The sky waves are the radio waves of frequency between 2 MHz to 30 MHz. These waves can propagate through atmosphere and are reflected back the ionosphere of the earth's atmosphere.

12. **Ans (1)**

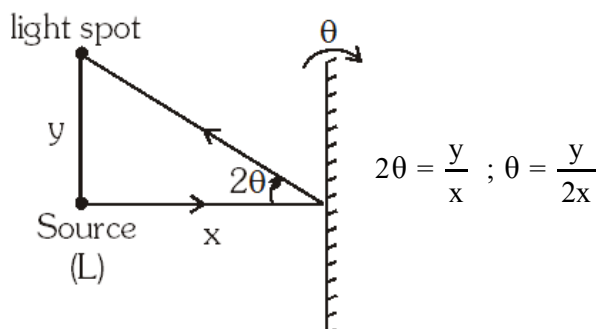


$$F = \int_0^L dm \omega^2 x \quad F = \int_0^L \left(\frac{M}{L} \right) dx \omega^2 x$$

$$F = \int_0^L \left(\frac{M}{L} \omega^2 \right) x dx = \frac{M}{L} \omega^2 \left[\frac{x^2}{2} \right]_0^L$$

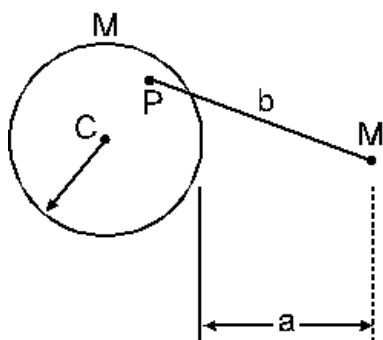
$$F = \frac{M \omega^2}{L} \frac{L^2}{2} \quad F = \frac{M \omega^2 L}{2}$$

13. **Ans (4)**



14. **Ans (4)**

At centre



$$V_c = -\frac{GM}{a} - \frac{GM}{2a}; \quad E_c = \frac{GM}{(2a)^2}$$

At any point P inside $V_P = -\frac{GM}{a} - \frac{GM}{b}$

$$E_P = \frac{GM}{b^2} \text{ \{only due to outside mass M\}}$$

15. **Ans (3)**

$$Q = \sigma A t (T^4 - T_0^4)$$

If T, T_0 , σ and t are same for both bodies then

$$\frac{Q_{\text{sphere}}}{Q_{\text{cube}}} = \frac{A_{\text{sphere}}}{A_{\text{cube}}} = \frac{4\pi r^2}{6a^2} \dots\dots(i)$$

But according to problem, volume of sphere

$$= \text{Volume of cube} \Rightarrow \frac{4}{3}\pi r^3 = a^3 \Rightarrow a = \left(\frac{4}{3}\pi\right)^{1/3} r$$

Substituting the value of a in equation (i) we get

$$\begin{aligned} \frac{Q_{\text{sphere}}}{Q_{\text{cube}}} &= \frac{4\pi r^2}{6a^2} = \frac{4\pi r^2}{6\left\{\left(\frac{4}{3}\pi\right)^{1/3} r\right\}^2} \\ &= \frac{4\pi r^2}{6\left(\frac{4}{3}\pi\right)^{2/3} r^2} = \left(\frac{\pi}{6}\right)^{1/3} : 1 \end{aligned}$$

16. **Ans (4)**

$$\gamma = 1 + \frac{2}{f}$$

17. **Ans (4)**

Quantity of gas in these bulbs is constant, i.e., Initial No. of moles in both bulb = Final number of moles

$$\mu_1 + \mu_2 = \mu_1 + \mu_2$$

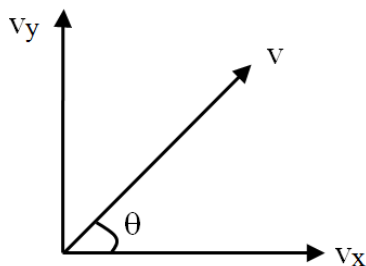
$$\frac{PV}{R(273)} + \frac{PV}{R(273)} = \frac{1.5PV}{R(273)} + \frac{1.5PV}{R(T)}$$

$$\Rightarrow = + \Rightarrow T = 819 \text{ K} = 546^\circ\text{C}$$

18. **Ans (1)**

$$\vec{v} = \frac{d\vec{r}}{dt} = 3\hat{i} + 4t\hat{j}$$

$$\text{At } t = 1\text{ s, } \vec{v} = (3\hat{i} + 4\hat{j}) \text{ m/s}$$



$$|\vec{v}| = \sqrt{3^2 + 4^2} = 5 \text{ m/s}$$

$$\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right)$$

19. **Ans (2)**

As the particle center region II follow circular path.

$$r = \frac{mv}{qB}$$

If the width of the region 1 is less than r the particle will enter region III

$$\frac{mv}{qB} > \ell$$

path length of the particle in region II will max

$$\text{when } \ell = r \quad v = \frac{q\ell B}{m}$$

$$\text{When } r = \ell \text{ time spent in the region II is } \frac{T}{2} = \frac{1}{2} \frac{2\pi m}{qB} = \frac{\pi m}{qB}$$

Time is independent of speed.

20. **Ans (3)**

de-Broglie wavelength of gas molecules

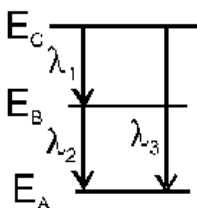
$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{3mRT}} \Rightarrow \frac{\lambda_H}{\lambda_{He}} = \sqrt{\frac{m_{He}T_{He}}{m_H T_H}} = \sqrt{\frac{4(273 + 127)}{2(273 + 27)}} = \sqrt{\frac{8}{3}}$$

21. **Ans (1)**

$$V_0 = \frac{h}{e}v - \frac{\phi}{e}$$

22. **Ans (2)**

$$E_C - E_A = (E_C - E_B) + (E_B - E_A)$$



$$\Rightarrow \frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$

$$\Rightarrow \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

23. **Ans (2)**

$$y = \frac{x^2}{20}$$

$$\frac{dy}{dx} = \frac{x}{10} = \tan \theta = \mu$$

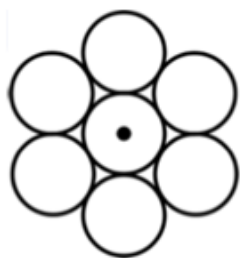
$$\frac{x}{10} = 0.5$$

$$\therefore x = 5\text{m}$$

$$\text{So height } y = \frac{x^2}{20} = \frac{25}{20}$$

$$y = 1.25 \text{ m}$$

24. **Ans (4)**



$$I = \frac{mr^2}{2} + 6 \left[\frac{mr^2}{2} + m(2r)^2 \right]$$

25. **Ans (2)**

$$\Delta R = \frac{\Delta V}{\Delta I} = \frac{1.2 - 1}{(15 - 10) \times 10^{-3}}$$

$$= \frac{0.2 \times 1000}{5} = \frac{200}{5} = 40\Omega$$

26. **Ans (3)**

NCERT XI ; Part-II ; Page No. 351

Amplitude of damped oscillation is

$$A = A_0 e^{-rt} \quad \text{at } t = 1 \text{ min} \quad A = \frac{A_0}{4}$$

$$\text{so } \frac{A_0}{4} = A_0 e^{-r} \text{ or } e^r = 4$$

$$\text{after 3 min } A = A_0 e^{-3r}$$

$$\Rightarrow \frac{1}{4} = (e^{-r})^3 \Rightarrow x = (4^3)$$

27. **Ans (1)**

$$\text{Beats per second} = n_1 - n_2 = 253 - 250 = 3$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2 = \left(\frac{4 + 2}{4 - 2} \right)^2 = 9$$

28. **Ans (3)**

$$\text{Dimensionally } F = \frac{x}{e}$$

$$\frac{M^1 L^1 T^{-2}}{1} = \frac{x}{M^1 L^{-3}}$$

$$\text{then } x = M^2 L^{-2} T^{-2}$$

$$\text{and } y = F$$

$$y = M^1 L^1 T^{-2}$$

29. **Ans (1)**

$$V = n\lambda$$

$$\lambda = \frac{V}{n} = \frac{10}{100} = 0.1$$

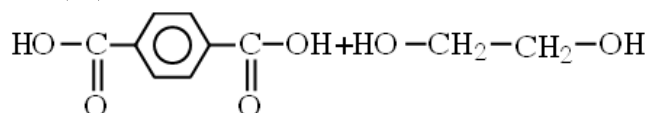
$$\Delta\phi = \frac{2\pi}{\lambda}, \Delta x = \frac{2\pi}{0.1} \times \frac{2.5}{100}$$

$$\Delta x = \frac{\pi}{2}$$

30. **Ans (3)**

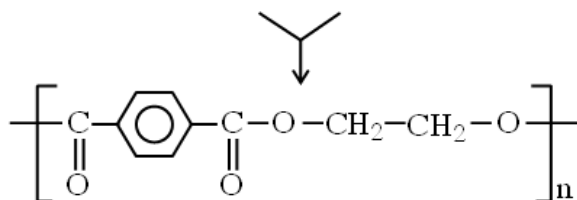
Diffraction always accompanies interference because it requires a single wave which interferes with itself. Thus both interference and diffraction occur.

31. **Ans (2)**



(Terephthalic acid)

(Ethylene glycol)



Dacron

32. **Ans (1)**

$$n_N = 20$$

$$n_{\text{CO}(\text{NH}_2)_2} = 10$$

$$\text{wt}_{\text{CO}(\text{NH}_2)_2} = 10 \times 60$$

$$= 600$$

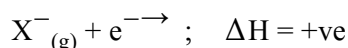
33. **Ans (2)**

Some drugs do not bind to the enzyme's active site. These bind to a different site of enzyme which is called allosteric site

34. **Ans (3)**

F^- ion concentration up to 1 ppm in drinking water is safe for teeth, but F^- ion concentration above 2 ppm is harmful for teeth and above 10 ppm is harmful for bones.

35. **Ans (1)**



36. **Ans (3)**

BOD values of clean water (A) is less than 5 ppm

So $A < 5$

BOD values of highly polluted water (B) is greater than 17 ppm

So $B > 17$

So Ans. is 3

37. **Ans (3)**

$$\Delta_t \approx \frac{4}{9} \Delta_0$$

$$\Delta_{[\text{CoCl}_4]^{2-}} \gg \frac{4}{9} \times 18000 = 8000 \text{ cm}^{-1}$$

38. **Ans (3)**

We have $b = 4N_A \left(\frac{4}{3} \pi r^3 \right)$. Thus

$$r = \left(\frac{3b}{16N_A \pi} \right)^{1/3} = \left[\frac{3 \times 25.12 \text{ cm}^3 \text{ mol}^{-1}}{16(6.022 \times 10^{23} \text{ mol}^{-1})(3.14)} \right]^{1/3}$$

$$\simeq \left(\frac{1}{4 \times 10^{23}} \text{ cm}^3 \right)^{1/3}$$

39. **Ans (3)**

The secondary structure of protein includes two type :

(a) α -Helix (b) β -pleated sheet

In α -Helix structure, the poly peptide chain is coil around due to presence of Intramolecular H-Bonding.

40. **Ans (1)**

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mKE}}$$

$$KE = qV \quad \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{KE_2}{KE_1}} = \sqrt{\frac{400}{100}} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{2}{1}$$

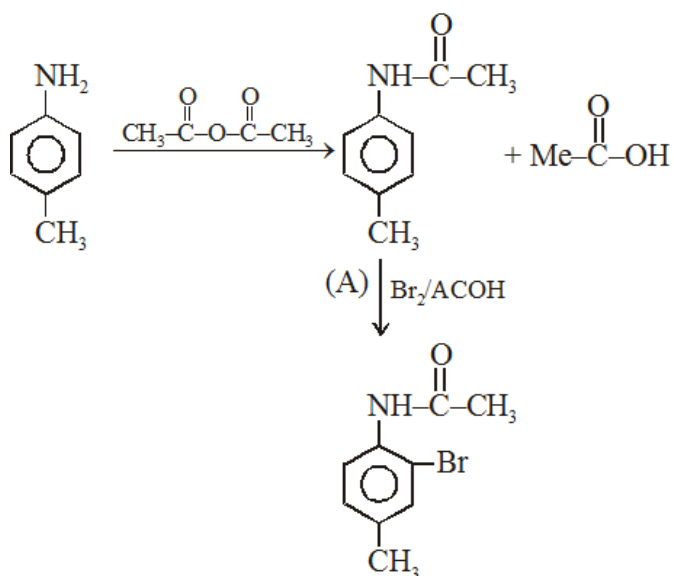
41. **Ans (2)**

$$\begin{aligned} \text{Rate} &= K[A]^2[B] = K[2A]^2[2B] \\ &= 8K[A]^2[B] \end{aligned}$$

42. **Ans (3)**

	Zn	Cr	Fe	Co	Cu
	↓	↓	↓	↓	↓
Density/g cm ⁻³	7.1	7.19	7.8	8.7	8.9

43. Ans (1)



44. Ans (2)

$$500 \times 2 = 0.5 \times V$$

$$\Rightarrow V = 2000 \text{ ml}$$

$$\text{added volume} = 2000 - 500 = 1500 \text{ ml}$$

45. Ans (2)

H₂O is neutral while alkali metal oxides are basic in nature.

46. Ans (2)

(A), (C) and (D) compounds do not have complete cyclic conjugation, so they are non aromatic.

47. Ans (3)

$$\text{pH} = 2 \Rightarrow [\text{H}^+] = 10^{-2} \text{ M} ; M_1$$

$$\text{pH} = 3 \Rightarrow [\text{H}^+] = 10^{-3} \text{ M} ; M_2$$

$$\text{Moles of HCl removed} = \underset{(\text{moles})}{n_{\text{H}^{\oplus}(\text{initial})}} - \underset{(\text{moles})}{n_{\text{H}^{\oplus}(\text{final})}}$$

$$= 10^{-2} \times 1 - 10^{-3} \times 1$$

$$= 9 \times 10^{-3} = 0.009$$

48. Ans (1)

Because of stability of half filled f-subshell configuration

49. Ans (2)

$$\frac{r_{\text{mix}}}{r_{\text{O}_2}} = \frac{\left(\frac{1}{311}\right)}{\left(\frac{2}{1200}\right)} = \sqrt{\frac{32}{M_{\text{mix}}}}$$

$$\frac{600}{311} = \sqrt{\frac{32}{M_{\text{mix}}}}$$

$$3.725 = \frac{32}{M_{\text{mix}}}$$

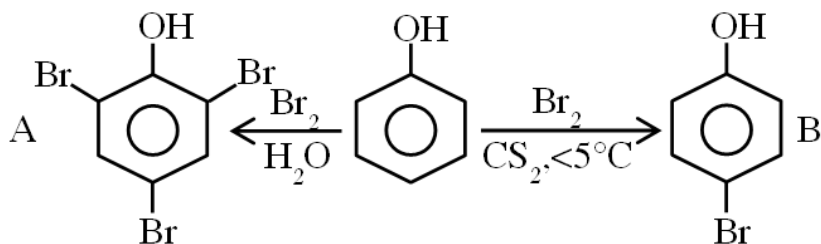
$$M_{\text{mix}} = 8.6$$

$$\text{V. D.} = \frac{8.6}{2} = 4.3$$

50. **Ans (3)**

In aluminothermite process Al acts as a reducing agent for Cr_2O_3 , Mn_3O_4 etc.

51. **Ans (2)**



52. **Ans (1)**

Generally, rate of $\text{S}_{\text{N}}2$ reaction $\propto$ Electrophilicity at the carbon where leaving group attached.

So, (1) (iii) > (ii) > (i)

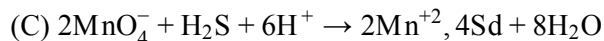
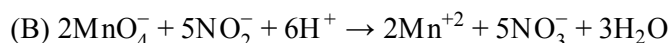
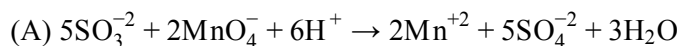
53. **Ans (4)**

The element E is Ga and the diagonal element of 5th period is $_{50}\text{Sn}$ having outer electronic configuration will be $[\text{Kr}] 5s^2 4d^{10} 5p^2$.

54. **Ans (2)**

Chiral centre present with symmetry in the molecule.

55. **Ans (4)**



56. **Ans (3)**

In $\text{A}_3(\text{BC}_4)_2$, $(+2) \times 3 + 2 [+5 + 4(-2)]$

$$\Rightarrow +6 + 10 - 16 = 0$$

Hence in the compound $\text{A}_3(\text{BC}_4)_2$, the oxidation no. of 'A', 'B' and 'C' are +2, +5 and -2 respectively.

57. **Ans (2)**

$$\frac{r_+}{r_-} = \frac{200}{500} = 0.4 \text{ (0.225 to 0.414) tetrahedral void}$$

58. **Ans (2)**

Due to low external pressure.

59. **Ans (1)**

$$W = -1 \times \frac{1+2}{2}$$

$$= -1.5 \text{ bar} \cdot \text{m}^3$$

$$= \frac{-1.5 \times 1000 \times 100}{1000} \text{ kJ}$$

$$= -150 \text{ kJ}$$

and $q = +100 \text{ kJ}$

$$\therefore \Delta U = q + w = -50 \text{ kJ}$$

60. **Ans (4)**

$$PE = 2TE$$

$$= 2 \left[-\frac{1}{2} \frac{kze^2}{r_n} \right]$$

$$r_n = 0.529 \frac{n^2}{Z} A^\circ = r_0 \frac{n^2}{Z} A^\circ$$

$$P.E. = - \frac{1 \times 3 \times e^2}{4\pi\epsilon_0 \times \frac{4r}{3}_o}$$

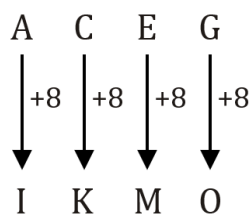
$$P.E = - \frac{9e^2}{16\pi\epsilon_0 r_0}$$

71. **Ans (3)**

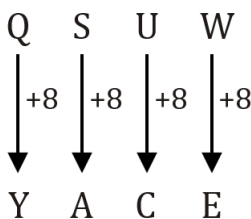
Meningitis is disease related to brain and cirrhosis is a disease related to liver.

72. **Ans (4)**

As,



Similarly,



73. **Ans (4)**

By observation,



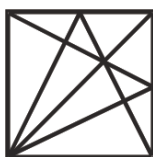
74. **Ans (1)**

By observation,



75. **Ans (4)**

By observation,



76. **Ans (2)**

All except, Argentina are continents, while Argentina is a country.

77. **Ans (3)**

In all other pairs, 2nd number = 1st number $\times$ 2.5

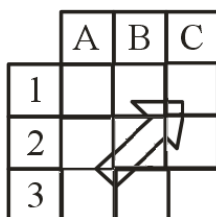
78. **Ans (3)**

All except option (3), we get same figure but in rotation form.

79. **Ans (4)**

By observation.

80. **Ans (2)**



81. **Ans (2)**

$$5^2 + 6^2 + 2^2 = 65 \div 5 = 13$$

$$5^2 + 8^2 + 4^2 = 105 \div 5 = 21$$

$$9^2 + 7^2 + 10^2 = 230 \div 5 = 46$$

82. **Ans (2)**

Putting A = 1, B = 3, C = 3, ..., M = 13,x X = 24, Y = 25, Z = 26, we have :

$$JR = \frac{J+R}{2} + 2 = \frac{10+18}{2} + 2 = 16$$

$$CK = \frac{C+K}{2} + 2 = \frac{3+11}{2} + 2 = 9$$

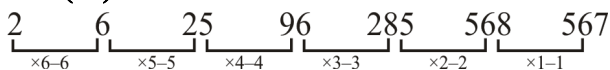
$$TX = \frac{T+X}{2} + 2 = \frac{20+24}{2} + 2 = 24$$

$$OS = \frac{O+S}{2} + 2 = \frac{15+19}{2} + 2 = 19$$

So, the missing numbers are :

21 and 14

83. **Ans (2)**

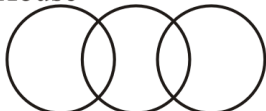


84. **Ans (3)**

Figures moves 90° CW and A.C.W rotation.

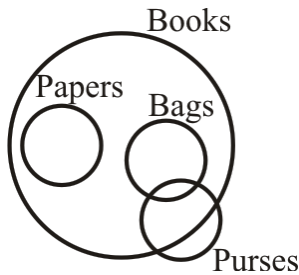
85. **Ans (4)**

House Office Schools



Hence, only II & IV and either I or III follows.

86. **Ans (2)**



87. **Ans (2)**

By row-wise observation, interior elements interchange their positions alternately.

88. **Ans (2)**

By observation,



89. **Ans (4)**

According to the passage, advanced countries are banning harmful asbestos products, rather than providing medical treatment to asbestos workers.

90. **Ans (2)**

According to the passage less than 1 per cent of 63,000 primary schools have all the facilities. Thus, less than 630 primary schools have the facilities.

91. **Ans (1)**

$$-\sqrt{20} \leq 2 \cos x - 4 \sin x \leq \sqrt{20}$$

$$\Rightarrow 0 \leq (2 \cos x - 4 \sin x)^2 \leq 20$$

$$\text{Min} = \frac{1}{1+20} = \frac{1}{21}; \quad \text{max.} = 1$$

$$\Rightarrow M + m = \frac{22}{21} \quad \text{Ans.]}$$

92. **Ans (1)**

$$f(x) = x e^{x(1-x)} \quad f'(x) = -e^{x(1-x)} (2x+1)(x-1) \quad f(x) \text{ is increasing in } \left(-\frac{1}{2}, 1 \right)$$

93. **Ans (1)**

$$\log_2(\log_3(\log_5(x))) = 0$$

$$\Rightarrow \log_3(\log_5(x)) = 2^0 = 1$$

$$\Rightarrow \log_5(x) = 3^1 = 3$$

$$\Rightarrow x = 5^3 = 125$$

94. **Ans (2)**

$$x|x| - 5|x+2| + 6 = 0$$

$$C-1 : x \in [0, \infty]$$

$$x^2 - 5x - 4 = 0$$

$$x = \frac{5 \pm \sqrt{25+16}}{2} = \frac{5 \pm \sqrt{41}}{2}$$

$$x = \frac{5 \pm \sqrt{41}}{2}$$

$$C-2 : x \in [-2, 0)$$

$$-x^2 - 5x - 4 = 0$$

$$x^2 + 5x + 4 = 0$$

$$x = -1, -4$$

$$x = -1$$

$$C-3 : x \in [-\infty, -2)$$

$$-x^2 + 5x + 16 = 0$$

$$x^2 - 5x - 16 = 0$$

$$x = \frac{5 \pm \sqrt{25+64}}{2}$$

$$\frac{5 \pm \sqrt{89}}{2}$$

$$x = \frac{5 - \sqrt{89}}{2}$$

95. **Ans (1)**

$$(1+x)^{41} (1-x+x^2)^{40} = (1+x) (1+x^3)^{40}$$

In the expansion of $(1+x^3)^{40}$, degree of all the terms will be multiple of 3. So no terms of degree 49 or 50.

96. **Ans (1)**

For continuity $f(1^-) = f(1^+)$

$$\Rightarrow a + 6 = b + 2 \Rightarrow a + 4 = b$$

and for differentiability $f'(1^-) = f'(1^+)$

$$\Rightarrow 2a + 6 = 2 \Rightarrow a = -2$$

97. **Ans (2)**

Putting $z = x + iy$ and $\arg(z_1/z_2) = \arg(z_1) - \arg(z_2)$,

we get

$$\tan^{-1} \frac{y}{x-1} - \tan^{-1} \frac{y}{x+1} = \frac{\pi}{4}$$

$$\text{or } \frac{[y/(x-1)] - [y/(x+1)]}{1 + [y^2/(x^2-1)]} = \tan \frac{\pi}{4} = 1$$

$$\text{or } 2y = x^2 + y^2 - 1$$

$$\text{or } x^2 + y^2 - 2y - 1 = 0 \Rightarrow \text{circle}$$

98. **Ans (3)**

$$\text{We have } z = \frac{1+7i}{(2-i)^2} = \frac{1+7i}{4+i^2-4i} = \frac{1+7i}{3-4i}$$

$$= \frac{(1+7i)(3+4i)}{(3-4i)(3+4i)} = \frac{-25+25i}{3^2+4^2} = -1+i$$

$$\therefore r = |z| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\text{Again, } z = -1+i, \therefore x = -1, y = 1$$

$$\text{Also, } \tan \theta = \left| \frac{y}{x} \right| = \left| \frac{1}{-1} \right| = 1,$$

$$\therefore \theta = \frac{\pi}{4}$$

$$\therefore x = -1 < 0 \text{ and } y = 1 > 0$$

Hence z lies in the 2nd quadrant.

$$\therefore \arg(z) = \pi - \theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\therefore \text{Polar form } z = \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

99. **Ans (3)**

Centres of the given circles are $C_1(1, -3)$ and $C_2\left(\frac{5}{2}, -3\right)$ and their radii are $r_1 = 2, r_2 = \frac{1}{2}$

$$\text{Here } C_1C_2 = r_1 - r_2$$

Hence circles touch internally.

100. **Ans (2)**

The two diameters intersect at $(8, -2)$ which is the centre of the circle. Then circle passes through $(6, 2)$

$$\therefore \text{Radius} = \sqrt{(8-6)^2 + (-2-2)^2} = \sqrt{20}$$

101. **Ans (3)**

$$\text{at } x = \frac{2\pi}{3}$$

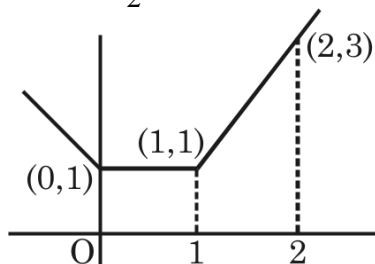
$$f(x) = \sin x - \cos x$$

$$\text{So } f'(x) = \cos x + \sin x$$

$$f'\left(\frac{2\pi}{3}\right) = \left(\frac{\sqrt{3}-1}{2}\right)$$

102. **Ans (3)**

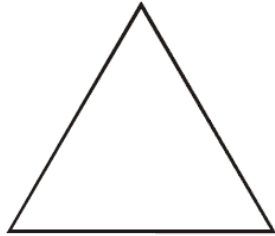
$$I = 1.1 + \frac{1}{2}(1+3).1 = 3$$



103. Ans (3)

$$\text{Area}(\Delta) = \begin{vmatrix} 2k+1 & k & 1 \\ 2 & 3 & 1 \\ k & 2k-1 & 1 \end{vmatrix} = 0$$

A(1, $\sqrt{3}$)



B $(-1, -\sqrt{3})$ C $(2\cos\theta, 2\sin\theta)$

$$\Rightarrow (2k+1)(3-2k+1) - k(2-k) + 1(4k-2-3k) = 0$$

$$\Rightarrow (2k+1)(4-2k) - k(2-k) + 1(k-2) = 0$$

$$\Rightarrow 8k - 4k^2 + 4 - 2k - 2k + k^2 + k - 2 = 0$$

$$\Rightarrow -3k + 5k + 2 = 0$$

$$\Rightarrow 3k^2 - 5k - 2 = 0$$

$$\Rightarrow 3k^2 - 6k + k - 2 = 0$$

$$\Rightarrow (3k+1)(k-2) = 0$$

$$k = \frac{-1}{3} \text{ and } k = 2$$

$$\text{so sum} = 2 - \frac{1}{3} = \frac{5}{3}$$

104. Ans (2)

$$\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = \begin{vmatrix} 5x-4 & 5x-4 & 5x-4 \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$= (5x-4) \begin{vmatrix} 1 & 1 & 1 \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix}$$

$$C_1 \rightarrow C_1 - C_2, \quad C_2 \rightarrow C_2 - C_3$$

$$= (5x-4) \begin{vmatrix} 0 & 0 & 1 \\ x+4 & -x-4 & 2x \\ 0 & x+4 & x-4 \end{vmatrix}$$

$$= (5x-4)(x+4)^2$$

$$\therefore (A, B) = (-4, 5)$$

105. Ans (3)

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$a^2 = 9, b^2 = 16$$

$$e^2 = 1 + \frac{b^2}{a^2} = \frac{25}{9}$$

$$e = \frac{5}{3}$$

Focus $(\pm 5, 0)$

since direction is $x = \frac{-9}{5}$

focus will be $(-5, 0)$

106. Ans (4)

Let $f(x_1) = f(x_2)$

$$\Rightarrow \frac{x_1^2 - 8}{x_1^2 + 2} = \frac{x_2^2 - 8}{x_2^2 + 2}$$

$$\Rightarrow (x_1^2 - 8)(x_2^2 + 2) = (x_2^2 - 8)(x_1^2 + 2)$$

$$\Rightarrow x_1^2 x_2^2 + 2x_1^2 - 8x_2^2 - 16 = x_1^2 x_2^2 - 8x_1^2 + 2x_2^2 - 16$$

$$\Rightarrow 10x_1^2 - 10x_2^2 = 0$$

$$\Rightarrow 10(x_1 - x_2)(x_1 + x_2) = 0$$

$$\Rightarrow x_1 - x_2 = 0 \text{ or } x_1 + x_2 = 0$$

$\therefore f$ is not one one

OR

$$f(x) = \frac{x^2 - 8}{x^2 + 2}$$

$$f(1) = f(-1)$$

(Q $f(x)$ is even function \ many one)

Range :

$$y = \frac{x^2 - 8}{x^2 + 2}$$

$$yx^2 + 2y = x^2 - 8$$

$$(y - 1)x^2 + 2y + 8 = 0$$

$$x^2 = \frac{2y + 8}{(1 - y)}$$

$$x = \pm \sqrt{\left(\frac{2y + 8}{1 - y}\right)}$$

$$\frac{2y + 8}{1 - y} \geq 0$$

$$\frac{y + 4}{y - 1} \leq 0$$

$$\begin{array}{c} + \quad - \quad + \\ | \quad | \quad | \\ -4 \quad 1 \\ \checkmark \quad \times \end{array}$$

$$\text{Range} = [-4, 1)$$

$\therefore$ Not onto

107. Ans (4)

$$\int \frac{1 + \tan^2 x}{1 - \tan^2 x} dx = \int \frac{\sec^2 x}{1 - \tan^2 x} dx$$

$$\text{Let } p = \tan x \quad dp = \sec^2 x \, dx$$

$$\int \frac{dp}{1 - p^2} = \frac{1}{2} \log \left| \frac{1 + p}{1 - p} \right| + c = \frac{1}{2} \log \left| \frac{1 + \tan x}{1 - \tan x} \right| + c$$

108. Ans (4)

$$\begin{aligned} & \int \left(\frac{x}{x \sin x + \cos x} \right)^2 dx = \\ & \int \left(\frac{x}{\cos x} \right) \cdot \frac{x \cos x \, dx}{(x \sin x + \cos x)^2} \\ & = \frac{x}{\cos x} \left(-\frac{1}{x \sin x + \cos x} \right) + \int \left(\frac{\cos x + x \sin x}{\cos^2 x} \right) \left(\frac{1}{x \sin x + \cos x} \right) dx \\ & = -\frac{x \sec x}{x \sin x + \cos x} + \int \sec^2 x \, dx \\ & = -\frac{x \sec x}{x \sin x + \cos x} + \tan x + C \end{aligned}$$

109. Ans (3)

$$f(x) = 4 \sin^{-1} \left(\frac{x^2}{1 + x^2} \right)$$

$$0 \leq \frac{x^2}{1 + x^2} < 1$$

$$\Rightarrow 0 \leq \sin^{-1} \left(\frac{x^2}{1 + x^2} \right) < \frac{\pi}{2}$$

$$\Rightarrow 0 \leq 4 \sin^{-1} \left(\frac{x^2}{1 + x^2} \right) < 2\pi$$

$$\text{Range : } [0, 2\pi)$$

110. Ans (2)

$$\begin{aligned} & \text{We have } \lim_{x \rightarrow 0} \left\{ \tan \left(\frac{\pi}{4} + x \right) \right\}^{1/x} \\ & = e^{\lim_{x \rightarrow 0} \left[\tan \left(\frac{\pi}{4} + x \right) - 1 \right] \cdot \frac{1}{x}} = e^{\lim_{x \rightarrow 0} \left\{ \frac{1 + \tan x}{1 - \tan x} - 1 \right\} \cdot \frac{1}{x}} \\ & = e^{\lim_{x \rightarrow 0} \frac{2 \tan x}{x(1 - \tan x)}} = e^{2 \lim_{x \rightarrow 0} \frac{\tan x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{(1 - \tan x)}} = e^{2 \cdot 1 \cdot 1} = e^2 \end{aligned}$$

111. Ans (3)

$$\begin{aligned} & \text{We have } \lim_{x \rightarrow 0} \frac{\sqrt{2} - \sqrt{1 + \cos x}}{\sin^2 x} \\ & = \lim_{x \rightarrow 0} \frac{2 - (1 + \cos x)}{\sin^2 x} \times \frac{1}{\sqrt{2} + \sqrt{1 + \cos x}} \\ & = \lim_{x \rightarrow 0} \frac{1 - \cos x}{(1 + \cos x)(1 - \cos x)} \times \lim_{x \rightarrow 0} \frac{1}{\sqrt{2} + \sqrt{1 + \cos x}} \\ & = \lim_{x \rightarrow 0} \frac{1}{(1 + \cos x)} \times \frac{1}{2\sqrt{2}} = \frac{1}{4\sqrt{2}} \end{aligned}$$

112. Ans (1)

$$\frac{dy}{dx} + \left(\frac{-x}{1-x^2} \right) y = \frac{1}{1-x^2}$$

$$\text{So, I.F.} = e^{-\int \frac{x dx}{1-x^2}} = \sqrt{1-x^2}$$

$$\text{Hence, } y\sqrt{1-x^2} = \int \frac{dx}{\sqrt{1-x^2}} + C$$

$$\Rightarrow y = (\sin^{-1}x + C)/\sqrt{1-x^2}$$

$$\text{Now, } c = 0 \text{ as } y(0) = 0$$

$$\therefore y = \frac{\sin^{-1}x}{\sqrt{1-x^2}}; y\left(\frac{1}{2}\right) = \frac{\pi/6}{\sqrt{3}/2} = \frac{\pi}{3\sqrt{3}}$$

113. Ans (1)

$$\text{Let } P = \begin{bmatrix} 2 & 1 \\ 7 & 4 \end{bmatrix}, Q = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}, R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$PAQ = R \Rightarrow A = P^{-1}RQ^{-1}$$

$$A = P^{-1}IQ^{-1} = P^{-1}Q^{-1} = \frac{1}{1} \begin{bmatrix} 4 & -1 \\ -7 & 2 \end{bmatrix} \frac{1}{(-1)} \begin{bmatrix} -3 & -2 \\ -5 & -3 \end{bmatrix} = \begin{bmatrix} 7 & 5 \\ -11 & -8 \end{bmatrix}$$

114. Ans (2)

Since A and B are independent

$$2/15 = P(A' \cap B) = P(A') P(B) = [1 - P(A)] P(B) \quad (1)$$

$$\text{and } 1/6 = P(A \cap B') = P(A) P(B') = P(A) [1 - P(B)] \quad (2)$$

Subtracting (2) from (1) we get

$$P(A) - P(B) = 1/30 \text{ or } P(A) = P(B) + 1/30$$

Putting this value in (2), we get

$$[P(B) + 1/30] [1 - P(B)] = 1/6$$

$$\Rightarrow 30P(B)^2 - 29P(B) + 4 = 0$$

$$\Rightarrow P(B) = 1/6, 4/5.$$

115. Ans (1)

X	P(x)	XP(X)	X ² P(X)
0	1/6	0	0
1	1/2	1/2	1/2
2	3/10	6/10	12/10
3	1/30	1/10	9/30

$$\sum xP(x) = \frac{6}{2} = \mu$$

$$\sigma^2 = \sum x^2P(x) - \mu^2$$

$$\sigma^2 + \mu^2 = 0 + \frac{1}{2} + \frac{12}{10} + \frac{9}{30} = 2$$

$$10(\sigma^2 + \mu^2) = 20$$

116. Ans (2)

Number of paths passing through M are

$$= \frac{7}{2 \ 5} \times \frac{4}{2 \ 2} = 126$$

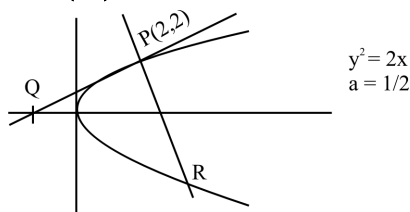
Number of paths through N are

$$= \frac{6}{2 \ 4} \times \frac{5}{2 \ 3} = 150$$

$$= 150 + 126 - 90 = 186$$

117. Ans (4)


$$\text{Number of such numbers} = {}^4C_3 \times \frac{3!}{2!} \times \frac{6!}{2!4!} = 180$$

118. Ans (1)


Tangent at P : $y(2) = 2(1/2)(x + 2)$

$$\Rightarrow 2y = x + 2$$

$$\therefore Q = (-2, 0)$$

$$\text{Normal at P : } y - 2 = -\frac{(2)}{2 \cdot 1/2}(x - 2)$$

$$\Rightarrow y - 2 = -2(x - 2)$$

$$\Rightarrow y = 6 - 2x$$

$$\therefore \text{Solving with } y^2 = 2x \Rightarrow R \left(\frac{9}{2}, -3 \right)$$

$$\begin{aligned} \therefore \text{Ar}(\Delta PQR) &= \frac{1}{2} \begin{vmatrix} 2 & 2 & 2 \\ -2 & 1 & 1 \\ \frac{9}{2} & -3 & 1 \end{vmatrix} \\ &= \frac{25}{2} \text{ sq. units} \end{aligned}$$

119. Ans (3)

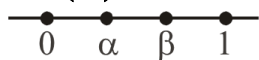
Let $p(h, k)$ be the point of contact

$$2y \frac{dy}{dx} = 4a \text{ \& } 2x = 4a \frac{dy}{dx}$$

both $\frac{dy}{dx}$ are same

$$\Rightarrow \frac{4a}{2y} = \frac{2x}{4a} \Rightarrow xy = 4a^2$$

120. Ans (2)



$$(a^2 - 1)x^2 + 2ax + 1 = 0$$

$$\therefore D = 4a^2 - 4(a^2 - 1) > 0$$

$$\therefore f(0) > 0, f(1) > 0, 0 < \alpha + \beta < 2$$

$$\Rightarrow a^2 - 1 + 2a + 1 > 0, 0 < \frac{-2a}{a^2 + 1} < 2$$

$$\Rightarrow a(a + 2) > 0, a < 0, a^2 + a + 1 > 0$$

$$\Rightarrow a < -2 \text{ Hence } a \in (-\infty, -2)$$

121. Ans (4)

Let a_n be the A.P. and first term is 'a' and common difference be 'd'

$$\therefore \text{Sum of } n \text{ terms} = S_n = \frac{n}{2} [2a + (n - 1)d]$$

Let b_n be the second A.P. and let first term be 'A' and common difference be 'D'

$$\therefore \text{Sum of } n \text{ terms} = S_n = \frac{n}{2} [2A + (n - 1)D]$$

$$\therefore \frac{S_a(n)}{S_b(n)} = \frac{5n + 9}{2n + 8}$$

$$\Rightarrow \frac{[2a + (n - 1)d]}{[2A + (n - 1)D]} = \frac{5n + 9}{2n + 8}$$

$$\Rightarrow \frac{a + \frac{(n-1)}{2}d}{A + \frac{(n-1)}{2}D} = \frac{5n + 9}{2n + 8}$$

$$\Rightarrow \text{put } n = 1$$

$$\Rightarrow \frac{a}{A} = \frac{14}{10} = \frac{7}{5}$$

$$\Rightarrow \frac{a_1}{b_1} = \frac{7}{5}$$

122. Ans (2)

$$417 = 17 + (n - 1)4$$

$$\Rightarrow 400 = 4(n - 1) \Rightarrow n = 101$$

$$\Rightarrow y \quad 466 = 16 + (m - 1)5$$

$$\Rightarrow 450 = 5(m - 1) \Rightarrow m = 91$$

Let T_n is common to both for some n for which m is an integer.

$$17 + (n - 1)4 = 16 + (m - 1)5$$

$$1 + 4n - 4 = 5m - 5$$

$$5m = 4n + 2$$

$$\text{hence } n = 2, 7, 12, \dots, 97 \rightarrow 20$$

123. Ans (2)

$$\text{Sum of 100 items} = 45 \times 100 = 4500,$$

$$\text{sum of correct obs.} = 19 + 31 = 50,$$

$$\text{sum of incorrect obser.} = 91 + 13 = 104$$

$$\text{corrected sum} = 4500 - 104 + 50 = 4446$$

$$\therefore \text{corrected mean} = \frac{4446}{100} = 44.46$$

124. Ans (1)

Equation of plane passing through (1,2,3) is given by

$$\frac{x-1}{a} = \frac{y-2}{b} = \frac{z-1}{c} \dots(1)$$

line (1) is parallel to plane $x - y + 2z = 5$

and $3x + y + z = 6$

$$\therefore a - b + 2c = 0$$

$$3a + b + c = 0$$

$$\frac{a}{-3} = \frac{b}{5} = \frac{c}{4}$$

required line

$$\frac{x-1}{-3} = \frac{y-2}{5} = \frac{z-3}{4}$$

125. Ans (4)

A vector in the direction of the required line can be obtained by cross product of

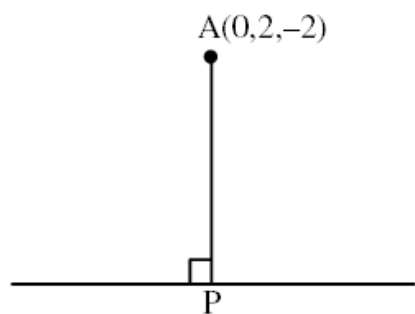
$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 5 \\ -1 & 3 & 2 \end{vmatrix} = -9\hat{i} - 9\hat{j} + 9\hat{k}$$

Required line,

$$\vec{r} = (5\hat{i} - 4\hat{j} + 3\hat{k}) + \lambda' (-9\hat{i} - 9\hat{j} + 9\hat{k})$$

$$\vec{r} = (5\hat{i} - 4\hat{j} + 3\hat{k}) + \lambda (\hat{i} + \hat{j} - \hat{k})$$

Now distance of (0, 2, -2)



$$\text{P.V. of P} \equiv (5 + \lambda)\hat{i} + (\lambda - 4)\hat{j} + (3 - \lambda)\hat{k}$$

$$\vec{AP} = (5 + \lambda)\hat{i} + (\lambda - 6)\hat{j} + (5 - \lambda)\hat{k}$$

$$\vec{AP} \cdot (\hat{i} + \hat{j} - \hat{k}) = 0$$

$$5 + \lambda + \lambda - 6 - 5 + \lambda = 0 ; \lambda = 2$$

$$|\vec{AP}| = \sqrt{49 + 16 + 9} ; |\vec{AP}| = \sqrt{74}$$

126. Ans (3)

$$\sin^7 x \leq \sin^2 x \leq 1 \quad \dots (1)$$

$$\text{and } \cos^7 x \leq \cos^2 x \leq 1 \quad \dots (2)$$

$$\text{also } \sin^2 x + \cos^2 x = 1$$

$\Rightarrow$ equality must hold for (1) & (2)

$$\Rightarrow \sin^7 x = \sin^2 x \text{ \& } \cos^7 x = \cos^2 x$$

$$\Rightarrow \sin x = 0 \text{ \& } \cos x = 1$$

or

$$\cos x = 0 \text{ \& } \sin x = 1$$

$$\Rightarrow x = 0, 2\pi, 4\pi, \frac{\pi}{2}, \frac{5\pi}{2}$$

$\Rightarrow$ 5 solutions

127. Ans (3)

$$(\cos 6^\circ \cdot \cos 54^\circ \cdot \cos 66^\circ)^2 + (\sin 6^\circ \cdot \sin 54^\circ \cdot \sin 66^\circ)^2$$

$$= \left(\frac{1}{4} \cos 18^\circ \right)^2 + \left(\frac{1}{4} \sin 18^\circ \right)^2$$

$$= \frac{1}{16} (\cos^2 18^\circ + \sin^2 18^\circ) = \frac{1}{16}$$

128. Ans (1)

$$\sin \left(x + \frac{11\pi}{27} \right) \cdot \cos \left(\frac{5\pi}{54} - x \right)$$

$$= \frac{1}{2} \left\{ \sin \frac{27\pi}{54} + \sin \left(2x + \frac{17\pi}{54} \right) \right\}$$

for minimum value

$$\left(2x + \frac{17\pi}{54} \right) = \frac{3\pi}{2}$$

$$\Rightarrow x = \frac{16\pi}{27}$$

129. Ans (2)

$\hat{a}, \hat{b}$ and $\hat{c}$ are unit vectors. Now

$$x = |\hat{a} - \hat{b}|^2 + |\hat{b} - \hat{c}|^2 + |\hat{c} - \hat{a}|^2$$

$$= 2 (\hat{a} \cdot \hat{a} + \hat{b} \cdot \hat{b} + \hat{c} \cdot \hat{c}) - 2\hat{a} \cdot \hat{b} - 2\hat{b} \cdot \hat{c} - 2\hat{c} \cdot \hat{a}$$

$$= 6 - 2 (\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}) \quad \dots (i)$$

$$\text{Also, } |\hat{a} + \hat{b} + \hat{c}| \geq 0$$

$$\text{or } \hat{a} \cdot \hat{a} + \hat{b} \cdot \hat{b} + \hat{c} \cdot \hat{c} + 2(\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}) \geq 0$$

$$\text{or } 3 + 2 (\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}) \geq 0$$

$$\text{or } 2 (\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a}) \geq -3$$

$$\text{from (1) } x \geq 9$$

130. Ans (1)

$$\vec{a} = 3\hat{i} + \hat{j}, \vec{b} = \hat{i} + 2\hat{j} + \hat{k}$$

$$\text{As } \vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda \vec{c}$$

$$\Rightarrow \vec{a} \cdot \vec{c} (\vec{b}) - (\vec{a} \cdot \vec{b}) \vec{c} = \vec{b} + \lambda \vec{c}$$

$$\Rightarrow \vec{a} \cdot \vec{c} = 1, \vec{a} \cdot \vec{b} = -\lambda$$

$$\Rightarrow (3\hat{i} + \hat{j}) \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -\lambda$$

$$\Rightarrow \lambda = -5$$